

LandWeb Fire-Model Inputs

The LandMine LTHFC (fire-return-interval) layer and its v10 update

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⚠ Warning

Draft report: pending human review

Portions of this document were drafted with the help of generative AI (large language models) and have not yet been fully reviewed by the co-authors. Content may be incomplete, inaccurate, or misattributed. This notice will be removed once a complete review has been carried out.

1 Overview

LandWeb simulates wildfire with an implementation of **LandMine** (Andison 1996; Andison 1998). The single most important spatial input to that fire model is the **LTHFC** (Long-Term Historic Fire Cycle) polygon layer: each polygon carries an integer **fire-return interval (FRI, years)** that sets how often that part of the landscape burns. This report documents the current **v10** LTHFC layer — in particular the **June 2026 Northwest Alberta (NWAB) update** (Peterson, Fermaniuk, Chubaty, et al. 2026; Peterson, Fermaniuk, Leverkus, et al. 2026) — and compares it against the previously used **v8c** layer and the **Spray Lake Sawmills (SLS v3)** variant.

The key message is a clean modelling seam: **v10 changes only the fire-return interval assigned to each zone. Everything else in the fire model is unchanged.** Any change in simulated fire regime attributable to inputs therefore traces to the FRI-by-zone values documented here.

2 How LandMine uses the LTHFC layer

The LTHFC polygons enter the pipeline as a fire-return-interval raster and drive ignition frequency:

- `LandWeb_preamble` downloads the v10 layer, renames its FRI field to `fireReturnInterval`, cleans the polygons (`LandWebUtils::polygonClean(type = "LandWeb", minFRI = ...)`), and rasterizes `fireReturnInterval` onto the study-area grid.
- `LandMine` consumes `fireReturnInterval` (years) together with a separate flammability mask (`rstFlammable`). The number of fires ignited per year in a zone is derived from that zone's area and its FRI (zone area / mean fire size / FRI).

Crucially, **only the fire-return interval comes from the LTHFC layer**. The fire-size distribution (a Truncated Pareto), the rate of spread (`R0STable`, `R0Sother`), and the per-pixel spread probability all come from LandMine's own parameters and its `DEoptim` calibration table — not from the LTHFC polygons. This is why a v8c-to-v10 swap is cleanly interpretable: it moves area between FRI classes and nothing else.

3 The v10 update: Northwest Alberta NRV (2026)

The v10 layer incorporates a major revision to the fire regime of **northwest Alberta**, produced by braiding Western science with Indigenous and long-time-resident knowledge (Peterson, Feramaniuk, Chubaty, et al. 2026; Peterson, Feramaniuk, Leverkus, et al. 2026). It replaces the original characterization from the LandWeb pre-industrial fire-regime synthesis (Andison 2019), which had assigned NW Alberta a **default 100-year** fire cycle — a value the synthesis itself flagged as lowest-confidence and likely too long.

3.1 Method (in brief)

1. **Time-since-fire** reconstruction from four stand-origin layers (reference years 1990 and 2023, each in *Liberal* and *Conservative* confidence in mapped fires), built from a hierarchical source-priority rule set (fire-history records, then undisturbed AVI, pre-harvest AVI, minimum harvest age, adjacency, and the 1949–1956 Broad-Scale Inventory).
2. **Fire cycle** estimated as the slope of cumulative undisturbed area against time-since-fire on a semi-log plot (the method of Reed et al. 1998, per Peterson, Feramaniuk, Chubaty, et al. 2026). An inflection near **1960** separates the pre-industrial regime (fire cycles of about 43–53 yr) from the fire-control era (about 354–460 yr).
3. **Three scenarios** bracket the estimate — *Shortest*, *Intermediate*, and *Longest* — cut to Canadian eco-districts so the zones integrate seamlessly into the larger LandWeb map.

The project recommends, and LandWeb adopts, the **Intermediate** scenario.

3.2 Fire-return intervals by natural subregion

Table 1: NWAB fire-return intervals (years) by Alberta natural subregion, per scenario. Reproduced from Peterson, Feramaniuk, Chubaty, et al. (2026), Table 2-5; the adopted Intermediate column is in bold.

Natural subregion	Shortest	Intermediate (adopted)	Longest
Boreal Subarctic	45	55	65
Central Mixedwood	45	55	65
Dry Mixedwood	35	40	45
Lower Boreal Highlands	45	55	65
Northern Mixedwood	45	55	65
Upper Boreal Highlands	35	45	55

Against the original 100-year default, the adopted intervals of **40–55 years** roughly *halve* the fire cycle across NW Alberta, which (per the Final Report) raises the modelled proportion of young forest and lowers old forest in the natural range of variation for that region.

4 Version comparison: v10 vs v8c vs SLS v3

This section is derived directly from the LTHFC shapefiles in `inputs/` by `R/lthfc_summary.R` (area-by-FRI over each layer). The three layers span slightly different extents and schemas: **v10** (3,002,148 km², field LTHFC10, LCC projection), **v8c** (2,905,726 km², field LTHFC, UTM 11N), and **SLS v3** (2,763,572 km², field LTHFC, UTM 11N).

4.1 Where the change is

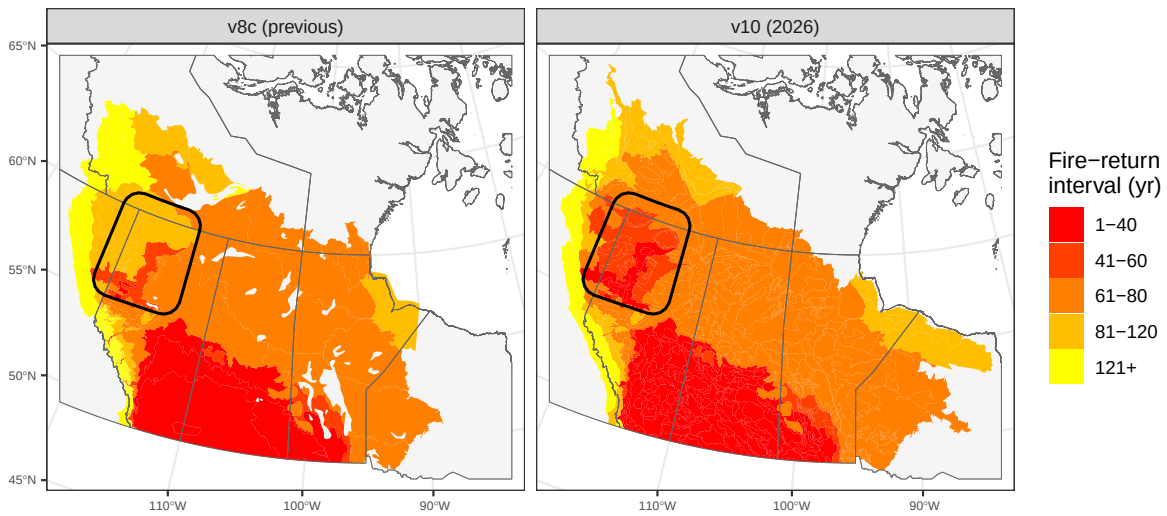


Figure 1: Fire-return interval across the LandWeb study area under v8c (previous) and v10 (2026). Colour runs red (short fire cycle) to yellow (long), following the NWAB mapping convention. The Northwest Alberta area revised in 2026 is outlined in black: inside it, v10 shifts toward red (shorter intervals), while the rest of the map is largely unchanged.

The two versions are near-identical outside Northwest Alberta; within the outlined AOI, v10 shifts toward red (shorter fire cycles). The difference map below isolates that change directly.

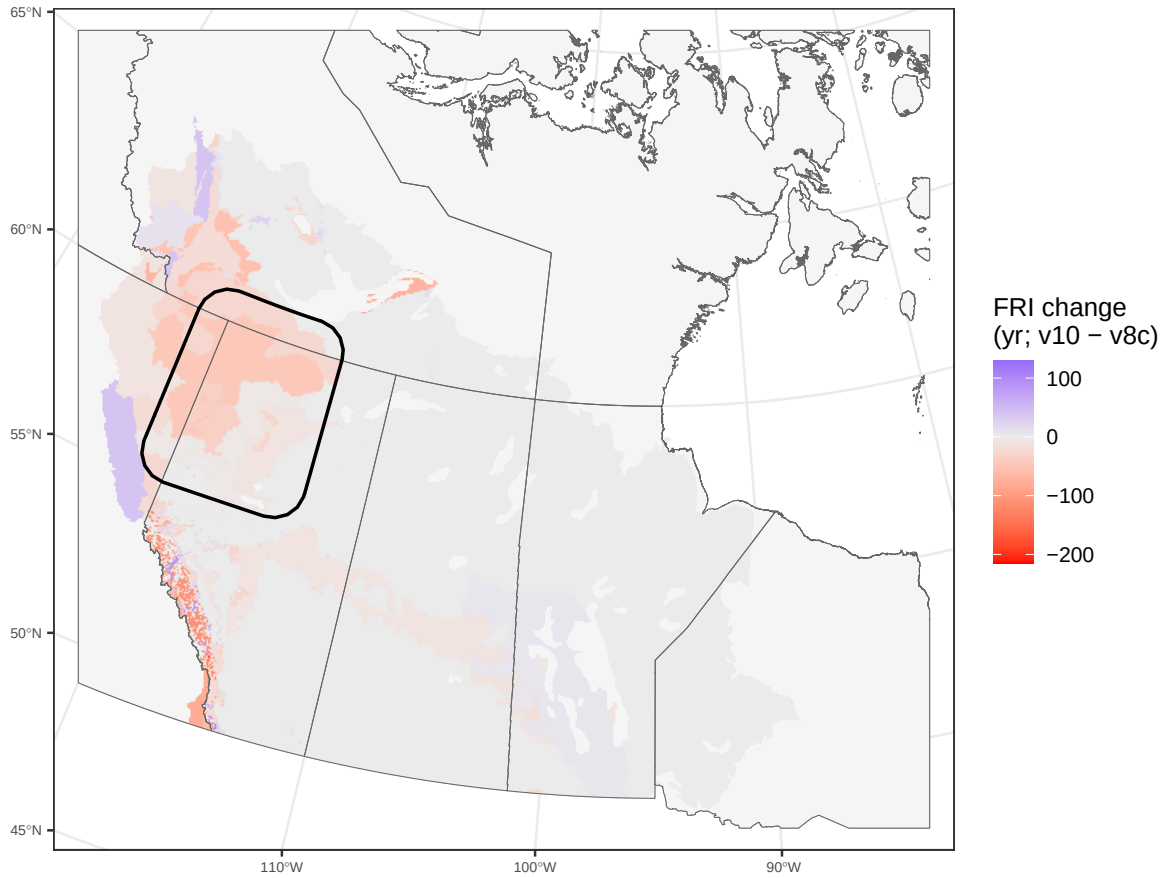


Figure 2: Change in fire-return interval, v10 minus v8c (years), over areas burnable in both versions. Red = v10 shortened the fire cycle (more fire); blue = lengthened; grey = unchanged. The change is concentrated in the outlined Northwest Alberta AOI, where the 2026 update roughly halved the cycle.

4.2 Area by fire-return interval

Table 2: Study-area area (thousand km²) by fire-return-interval band (years) under v8c vs v10 (burnable polygons only).

FRI band (yr)	v8c (1000 km ²)	v10 (1000 km ²)
1-40	564.5	633.5
41-60	110.1	210.7
61-80	1398.5	1559.4
81-120	448.5	406.5
121+	264.1	192.1

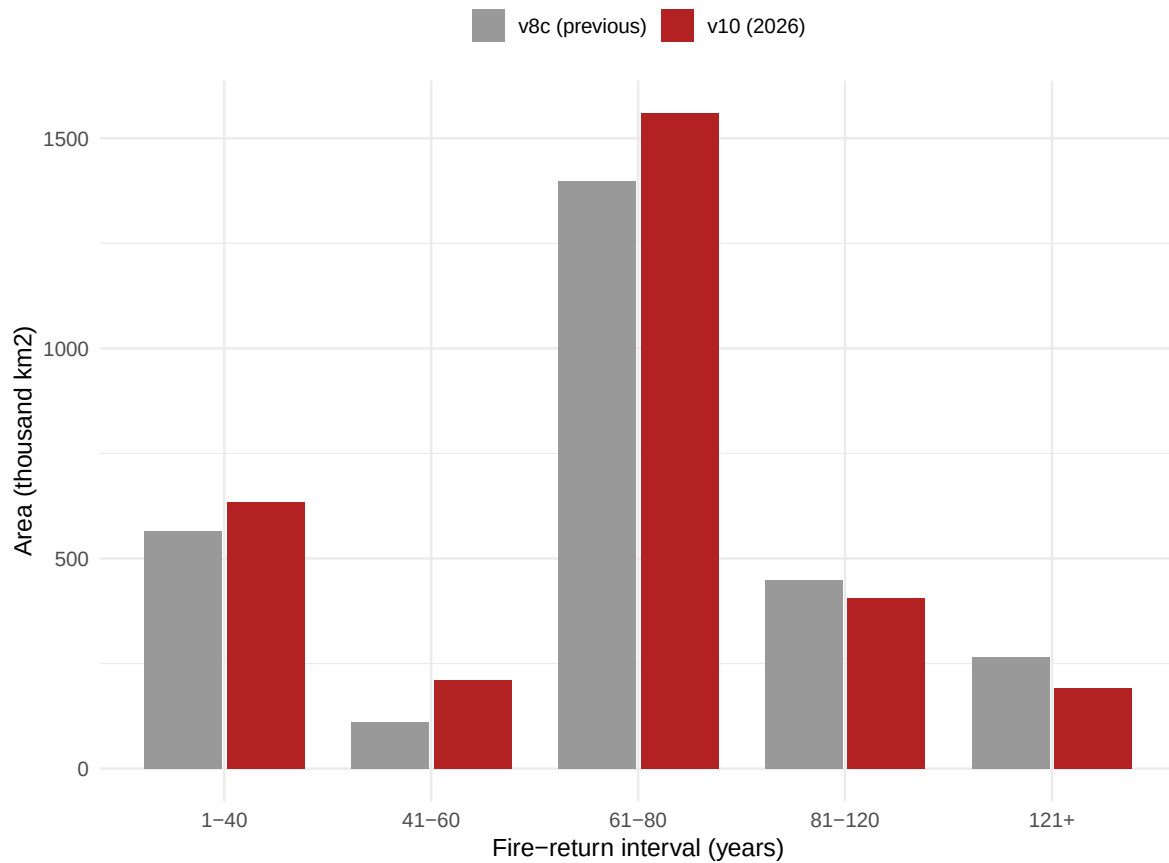


Figure 3: Study-area area by fire-return-interval band, v8c vs v10 (burnable polygons only). v10 shifts area out of the longest intervals and into the 41–60 yr band — the NW Alberta halving.

The signature of the NW Alberta update is visible at the landscape scale: relative to v8c, v10 moves roughly **101 thousand km²** of additional area into the **41–60 year** band (110 to 211

thousand km²), drawing it down from the **81-120** and **over-120** year bands. Note this is a *partial* update: v10 revises NW Alberta only, so most of the map retains its earlier intervals, and full re-integration of the Intermediate scenario across the whole LandWeb LTHFC map remains outstanding (Peterson, Fermaniuk, Leverkus, et al. 2026).

4.3 The schema and non-burnable-coding change

The v10 layer is not just new values — it is a new schema. The FRI field was renamed (LTHFC to LTFC10), the projection changed (UTM 11N to LCC), and non-burnable land is handled differently: **v8c encoded 119,953 km² as fire-return interval 0**, whereas v10 carries no 0 class and instead marks water via a lakes attribute. Downstream, `minFRI` in the preamble still maps very-short intervals to non-burnable; with v10's 35-year minimum nothing is dropped at the pipeline's current `minFRI = 25`.

The **SLS v3** layer is the Spray Lake Sawmills variant of the map (a near-full-extent layer, not a local patch); it was the source from which the Spray Lake / Crowsnest polygons were taken to build the interim **v8d** patch (`R/new_lthfc.R`). It is included here for the requested comparison but is not the current pipeline input.

5 Expected effect on the simulated fire regime

Because LandMine derives ignition frequency directly from FRI, shortening NW Alberta's fire cycle increases the number of fires there, which over a multi-century run raises the proportion of young/recently-burned forest and lowers old forest — shifting the seral-stage natural range of variation for that region. Quantifying that shift in simulated outputs (v2 vs v3) is the subject of the output-comparison report.

References

- Andison, David W. 1996. “Managing for Landscape Patterns in the Sub-Boreal Forests of British Columbia.” PhD Thesis, University of British Columbia. <https://open.library.ubc.ca/cIRcle/collections/ubctheses/831/items/1.0075275>.
- Andison, David W. 2019. *Pre-Industrial Fire Regimes of the Western Boreal Forest of Canada*. Synthesis Report. fRI Research.
- Andison, David W. 1998. “Temporal Patterns of Age-Class Distributions on Foothills Landscapes in Alberta.” *Ecography* 21 (5): 543–50. <https://doi.org/10.1111/j.1600-0587.1998.tb00446.x>.
- Peterson, E., C. Fermaniuk, A. M. Chubaty, S. Leverkus, and D. W. Andison. 2026. *Northwest Alberta Natural Range of Variation — Data Report: Braiding a Long-Time Understanding of Historical Fire Regimes in Northwest Alberta into LandWeb*. Forest Resource Improvement Association of Alberta (FRIAA).
- Peterson, E., C. Fermaniuk, S. Leverkus, A. Jorgensen, A. M. Chubaty, and D. W. Andison. 2026. *Northwest Alberta Natural Range of Variation — Final Report: Braiding a Long-Time Understanding of Historical Fire Regimes in Northwest Alberta into LandWeb*. Forest Resource Improvement Association of Alberta (FRIAA).

Provenance

- **v10 source:** the NWAB Intermediate fire-return intervals (Peterson, Fermaniuk, Chubaty, et al. 2026), cut into the LandWeb LTHFC map; `inputs/landweb_ltfc_v10.shp` (field LTHFC10). Original fire-cycle characterization from Andison (2019).
- **Comparison layers:** `inputs/landweb_ltfc_v8c.shp` (field LTHFC), `inputs/ltfc_sls_v3.shp` (field LTHFC).
- **Data file:** `outputs/extended_analyses/lthfc_version_summary.csv`
- **Method:** `R/lthfc_summary.R`, built by the gated `lthfc_version_summary_csv` target. Areas are geodesic (each layer reprojected to EPSG:4326, then `sf::st_area`) so the differently-projected layers are comparable. The scan is light and cached as a `format = "file"` target; delete the CSV to recompute.
- **FRI-by-NSR table** reproduced from Peterson, Fermaniuk, Chubaty, et al. (2026), Table 2-5.

Appendix: Build provenance

This appendix pins the rendered PDF to the code, environment, and revisions that produced it, so a result can be reproduced or a difference attributed to code vs. inputs vs. randomness. It is generated at render time from the `workflowtools info_*` family.

Build identity. LandWeb repository, branch `targets`, commit `ad683998`, rendered 2026-07-03 10:15 MDT with R version 4.6.1 (2026-06-24) on Ubuntu 26.04 LTS (GEOS 3.14.1, GDAL 3.12.2, PROJ 9.7.1).

Table 3: SpaDES modules and owned packages at their pipeline-pinned revisions when this report was rendered (`workflowtools::info_submodules()`).

Component	Kind	Pinned SHA
Biomass_borealDataPrep	module	f570094ea9
Biomass_core	module	32a3790870
Biomass_regeneration	module	b739da2d32
Biomass_speciesData	module	101ad29571
Biomass_speciesFactorial	module	01d4559d15
Biomass_speciesParameters	module	1e381a7881
HSI_Caribou_MB	module	1180d61743
LandMine	module	d7ccb083e1
LandWeb_output	module	af78cee648
LandWeb_preamble	module	65dc365f37
LandWeb_summary	module	99e229e3ac
NRV_summary	module	ca7b444fa2
burnSummaries	module	add6cc0bb4
timeSinceFire	module	30d242ea7b
LandWebUtils	package	cf29ea06e4
SpaDES.targets	package	b1f68a5467
crew.ssh	package	79c368f0fa

nrvttools	package	f9665fd7b0
spatialutils	package	a891aef9d5
workflowtools	package	561def4389

The full `sessionInfo()` and the exact `renv.lock` package pins accompany the run; input datasets are recorded in `_input_manifest.json` (report 06).

DRAFT